RAFFLES INSTITUTION

YEAR 56 PHYSICS DEPARTMENT

Tutorial 16 Circuits Suggested Solutions

D1 **(a)** **(i)** The applied potential difference of 15.4V has exceeded the maximum potential difference across the car bulb, 12V. As such, the filament breaks and the circuit is open.

(ii) At 4.0V, current is 3.0A. Resistance of headlamp $= \frac{4.0}{3.0} = 1.33\ \Omega$

At 12.0V, current is 5.5. Resistance of headlamp $= \frac{12.0}{5.5} = 2.18\ \Omega$

(iii) The resistance of a conductor is defined as the ratio of the potential difference across it to the current flowing through it (V/I), whereas the reciprocal of the gradient of the graph is the change in potential difference divided by the change in current (dV/dI). These quantities are not equal in general, and are equal only if the I-V characteristic graph is proportional (i.e. linear and passes the origin). The current I-V curve is not proportional; hence the two quantities are not equal.

(b) **(i)**

$$R = \rho \frac{l}{A}$$

$$l = \frac{AR}{\rho} = \frac{(\pi)(\frac{0.084\times10^{-3}}{2})^2(0.40)}{5.5\times10^{-8}} = 0.040\text{ m}$$

(ii) A straight length wire will occupy too much space and also have too much surface area exposed to convection currents.

As the wire is 4.0 cm, the lamp has to be bigger than 4.0cm which means light energy is spread over a much bigger surface area, the lamp may thus not give satisfactory intensity of light.

(iii) To produce a shorter filament while keeping resistance constant,

1. wire can be coiled, or

use a thinner wire so that a shorter length is required (assuming same R).

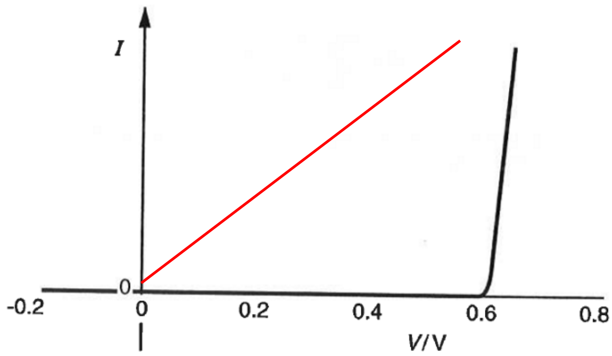
Examiner's report

(a) In **(i)**, few candidates realised that for a 12 V car bulb with 15.4 V across it the most likely problem is that the filament breaks. Answers such as 'the resistance of the headlamp will not reach zero' were common. Parts **(ii)** and **(iii)** were answered well.

(b) Most candidates coped well with **(i)** but when the answer appeared as 0.0403 m candidates did not visualise this as 4.0 cm. Those who did had no problem in suggesting that the wire be coiled or that a thinner wire be used in **(iii)**.

D2 **(a)** **(i)** When potential difference < 0.6 V, the current I through the diode is zero. Hence resistance of the diode is infinite. As potential difference increases from 0.6 V onwards, I increases sharply, e.g. I at 0.63 V is much higher than I at 0.61 V even though V increases by little, hence the ratio V/I i.e. resistance decreases with V.

(ii)



(b) **(i)** Volume $V = AL$

$$R = \rho\frac{L}{A} = \rho\frac{L}{V/L} = \rho\frac{L^2}{V}$$

Given that volume of the wire and its resistivity both remain constant, resistance is directly proportional to the square of its length.

(ii) $R \propto L^2$

$$\frac{R_f}{R_i} = \left(\frac{L_f}{L_i}\right)^2 = \left(\frac{\frac{105.2}{100}L_i}{L_i}\right)^2 = (1.052)^2 = 1.11\ (3\text{s.f.})$$

Examiner's report

(a) **(a)(i)** Marks were often lost for this part of the question because the candidates tried to explain why the resistance of the diode changed rather than use the graph to describe how the value changed with the potential difference applied across it. A significant number of candidates thought that the resistance was zero for low values of potential difference, even supporting their description with the detail that the current was negligible.

(b) **(b)(i)** The definition of resistivity was well-known but only the better candidates were able to use the information that the volume remained constant to go on to show how the resistance was related to the length.

(ii) This calculation was generally well done with only the weaker candidates unable to use the given relationship between resistance and length to calculate the correct ratio.

D3 **(a)** **(i)** $P = IV$

$9.0 = I(30)$

$I = 0.30$ A

(ii) $V = \frac{30}{12} = 2.5$ V

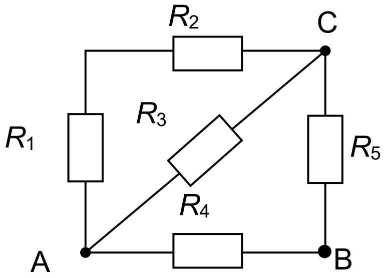
(iii) $R = \frac{V}{I} = \frac{2.5}{0.30} = 8.3\ \Omega$

(b) **(i)** The power supply is faulty and is not supplying a p.d. across the circuit, or wire between M and power supply (or that between A and the power supply) is broken.

(ii) The filament of the lamp between E and F is broken.

(iii) One (no need 2 or more) of the wires between A and M is broken, or filament of more than one lamp is broken.

D4 **(a)**



From the above circuit diagram:

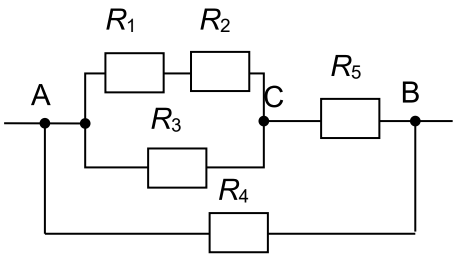
Equivalent resistor R_{AC}

- R_1 and R_2 are in series between A and C;
- R_3 is between A and C;

R_{AC} and R_5 are in series between A and B;

R_4 is between A and B;

Hence, the equivalent circuit is

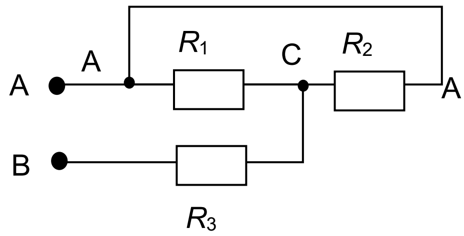


$$\frac{1}{R_{AC}} = \frac{1}{2R} + \frac{1}{R} \quad \Rightarrow \quad R_{AC} = \frac{2R}{3}$$

$$\frac{1}{R_{AB}} = \frac{1}{\frac{2}{3}R + R} + \frac{1}{R} = \frac{1}{\frac{5}{3}R} + \frac{1}{R} = \frac{8}{5R}$$

$$\Rightarrow R_{AB} = \frac{5}{8}R = 0.625R$$

(b)



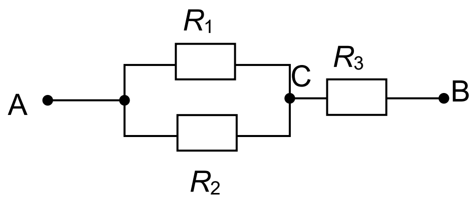
From the above circuit diagram:

R_1 is between A and C;

R_2 is between A and C;

R_3 is between B and C.

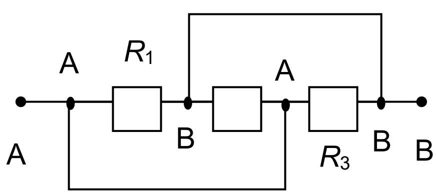
Hence, the equivalent circuit is



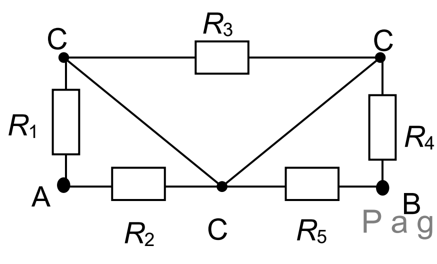
$$\frac{1}{R_{AC}} = \frac{1}{R} + \frac{1}{R}, \; R_{AC} = 0.5R$$

$R_{AB} = 0.5R + R = 1.5R$

(c)

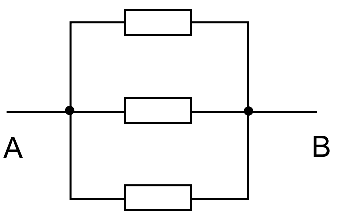


(d)



From the above circuit diagram:
R_1 is between A and B;
R_2 is between A and B;
R_3 is between A and B.

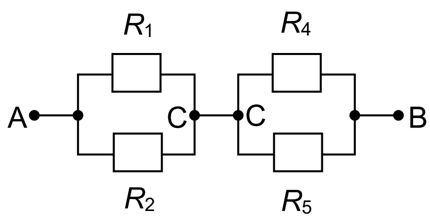
Hence the equivalent circuit is



$$R_{AB} = \frac{1}{3}R = 0.333R$$

From the above circuit diagram:
R_3 is between C and C i.e. $\Delta V_{R3} = 0$ => Remove
Both R_1 and R_2 are between A and C;
Both R_4 and R_5 are between C and B;

Hence, the equivalent circuit is:



$$R_{AB} = \left(\frac{1}{R} + \frac{1}{R}\right)^{-1} \times 2 = R$$

D5 **(a)** **(i)** When S is at Y, p.d. across XY = $\frac{20}{20 + 4.0} \times 12 = 10$ V

When S is at X, V_{XS} = 0 V and when S is at Y, V_{XS} = 10 V.

Voltage range is from 0 V to 10 V and S is negative with respect to X.

(ii) When S is at Y, p.d. across YZ = 12 – 10 = 2 V
When S is at X, V_{SZ} = 12 V. When S is at Y, V_{SZ} = 2 V.

Voltage range is from 2 V to 12 V and S is positive with respect to Z.

(b) The 10 kΩ resistance of the resistance track is in parallel with the 10 kΩ voltmeter. Hence, their combined resistance R_{SY} = 5.0 kΩ.

$$V_{SY} = \left(\frac{R_{SY}}{R_{XS} + R_{SY} + R_{YZ}}\right)V_{XZ} = \left(\frac{5.0}{10 + 5.0 + 4.0}\right) \times 12 = 3.2 \text{ V}$$

D6 **(a)** The internal resistance can be considered to be negligible as it is only ~ 0.0066% of the total resistance of a circuit (at 20°C) in which all circuit components are in series.

Note: There are two (equivalent) criteria to determine if a resistor is negligible in an arrangement:

1. If, upon removing the resistor, the effective resistance of the arrangement changes little, then the removed resistor is negligible;
2. If the power dissipated in a resistor is negligible compared to that in the other resistors, then the resistor is negligible.

According to either of these criteria, a resistor with small resistance is negligible in the series arrangement, whereas a resistor with large resistance is negligible in the parallel arrangement.

(b) **(i)** $V = \dfrac{4000}{4000+2000} \times 3.0 = 2.0\ V$

(ii) $V = \dfrac{1800}{1800+2000} \times 3.0 = 1.4\ V$

(c) At 0°C,

$\dfrac{R}{4000+R} \times 3.0 = 1.2\text{ V} \quad \Rightarrow \quad R = 2700\ \Omega$

At 20°C,

$\dfrac{R}{1800+R} \times 3.0 = 2.4\text{ V} \quad \Rightarrow \quad R = 7200\ \Omega$

There is no single value of *R* that will satisfy both conditions.

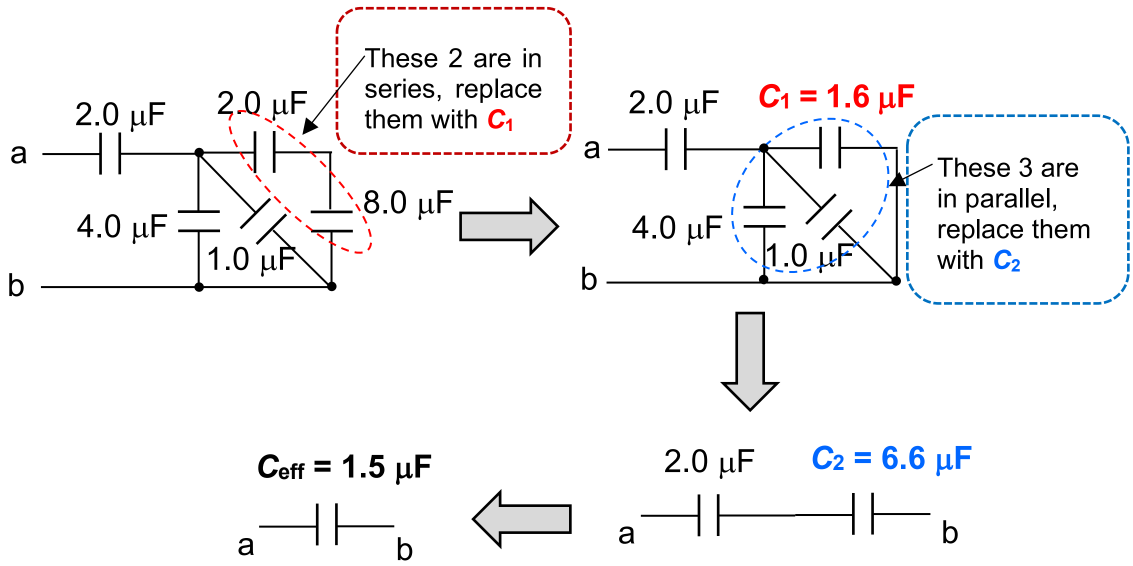
Examiner's report

(a) Candidates should be advised that a comparison is required in their answers. A statement that 'the resistance of the internal resistors is small' is insufficient and is almost a paraphrase of the question.

(b) These calculations presented very few problems.

(c) There were some very well explained answers where the relevant calculations were made alongside a summary comment. It was unfortunate that many candidates correctly calculated the resistance of the fixed resistor at the two temperatures but did not then make any comment on their answers.

D7 **(a)**



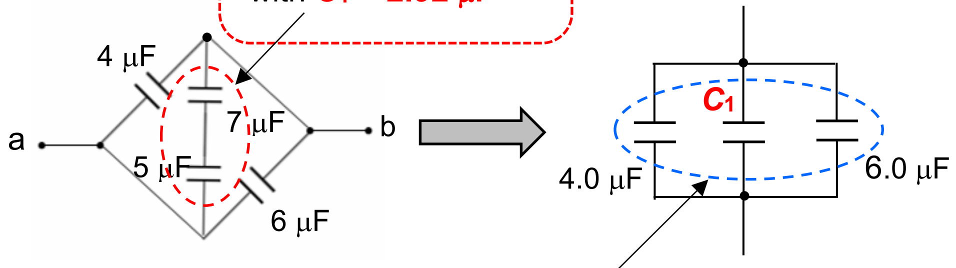
$\frac{1}{C_1} = \frac{1}{2.0} + \frac{1}{8.0} \Rightarrow C_1 = 1.6\ \mu F$

$C_2 = 4.0 + 1.0 + 1.6 = 6.6\ \mu F$

$\frac{1}{C_{eff}} = \frac{1}{2.0} + \frac{1}{6.6} \Rightarrow C_{eff} = 1.5\ \mu F$

(b)

These 2 are in series, replace them with C_1 = 2.92 μF



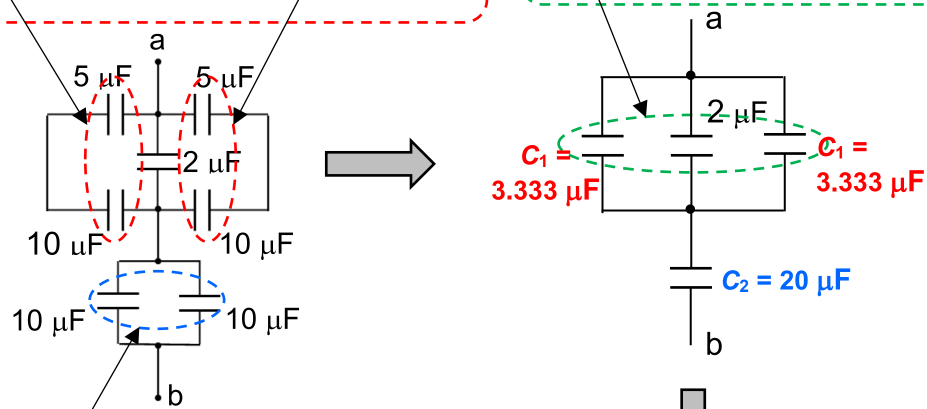
These 3 are in parallel.

$\therefore$ C_{eff} = 4.0 + 2.92 + 6.0 = 12.92 ≅ **13 μF**

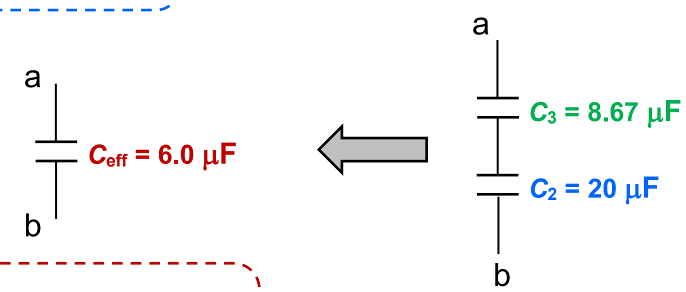
(c)

The 5 μF and the 10 μF are in series. Replace them with C_1 = 3.333 μF

These 3 are in parallel. Replace them with C_3 = 8.67 μF



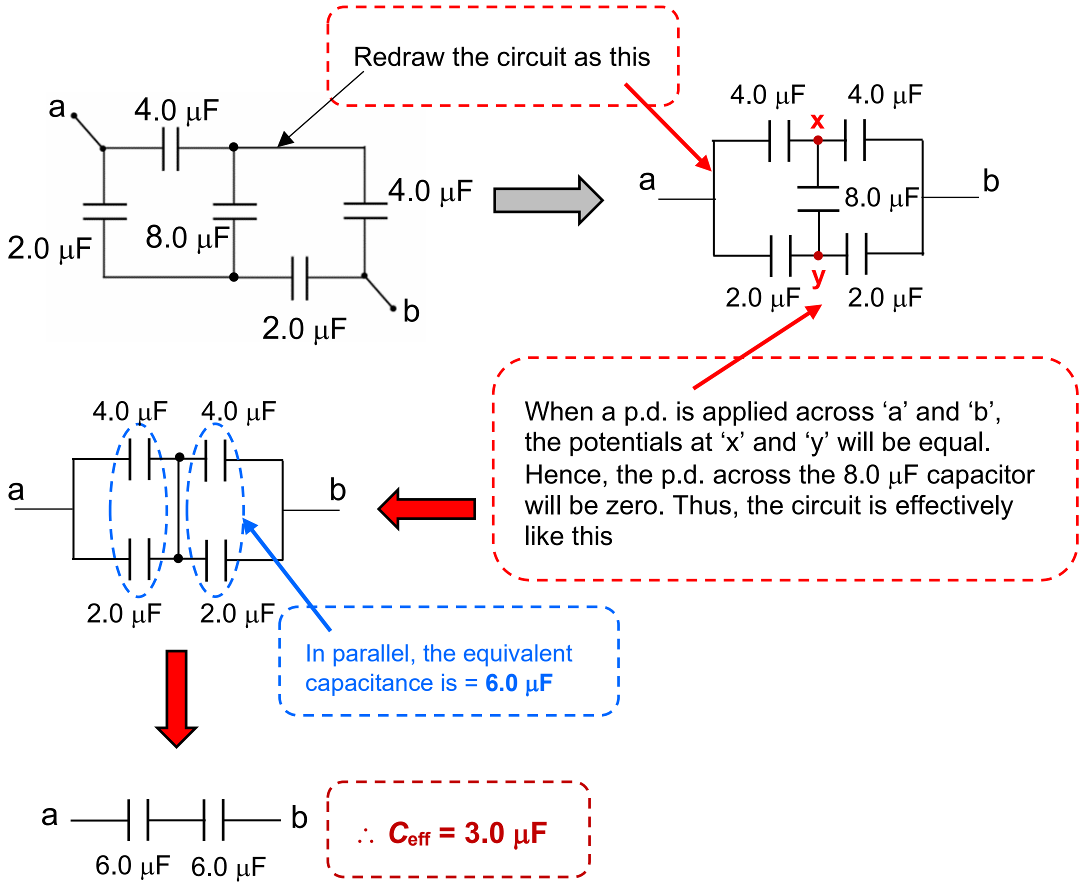
These 2 are in parallel. Replace them with C_2 = 20 μF



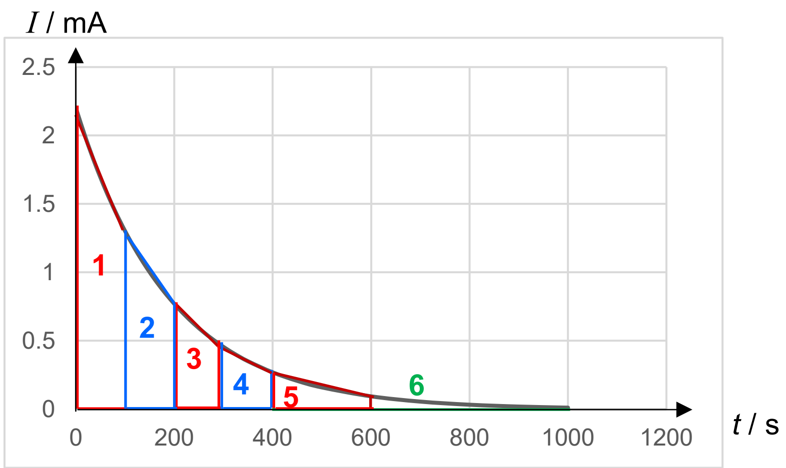
C_2 and C_3 are in series.

$\therefore$ **C_{eff} = 6.0 μF**

(d)



D8 **(a)**



Equilibrium charge Q_0 on capacitor = Total area under the current-time graph

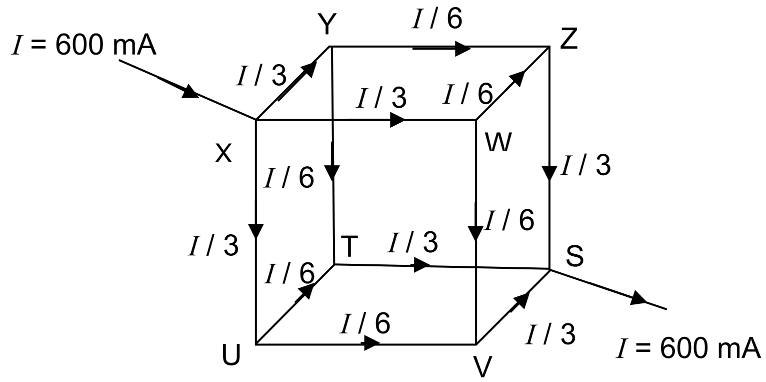
$= \frac{1}{2}(2.2 + 1.3)\times 100 + \frac{1}{2}(1.3 + 0.8)\times 100 + \frac{1}{2}(0.8 + 0.5)\times 100 +$

$\frac{1}{2}(0.5 + 0.3)\times 100 + \frac{1}{2}(0.25 + 0.1)\times 200 + \frac{1}{2}(0.1)\times 400 = 0.44\ \text{C}$

(b) $C = \dfrac{Q_0}{V} = \dfrac{0.44}{15} = 0.0293\,\text{F} \approx 0.029\,\text{F}$

(c) $R = \dfrac{V}{I_0} = \dfrac{15}{2.2\times10^{-3}} = 6818\ \Omega \approx 6800\ \Omega$

C1



(a) $R = \rho\dfrac{l}{A} = \left(4.9\times10^{-7}\right)\left(\dfrac{20\times10^{-3}}{0.66\times10^{-6}}\right) = 0.0148 = 1.5\times10^{-2}\Omega$

(b) Current in wires XY, XW, XU, ZS, TS, VS = I / 3 = 600 / 3 = **200 mA**

Current in wires YZ, YT, UT, UV, WZ, WV = I / 6 = 600 / 6 = **100 mA**

(c) p.d. between X and S = p.d. across wire with current I/3

+ p.d. across wire with current I / 6

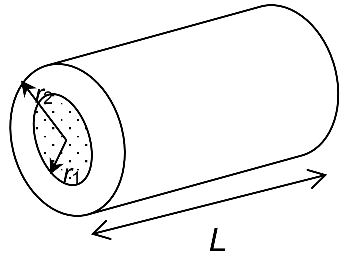
+ p.d. across wire with current I / 3

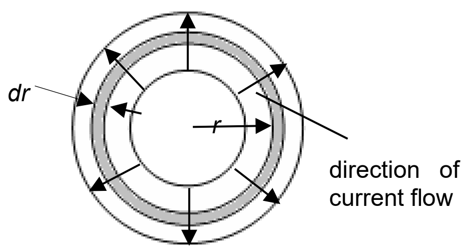
= 2 (1.5 x 10^{-2}) (100 x 10^{-3}) + (1.5 x 10^{-2}) (100 x 10^{-3})

= **7.5 x 10^{-3} V**

Resistance of network = $\dfrac{7.5\times10^{-3}}{600\times10^{-3}} = 0.0125 = \mathbf{1.3\times10^{-2}}\ \Omega$

C2





end view of hollow cylinder

Since current flow is perpendicular to the cylinder's axis, the thickness of the material through which the current flows through changes as radius changes.

Divide the cylinder into concentric elements of hollow cylinders each of infinitesimal thickness dr.

From $R = \rho\dfrac{l}{A}$, the resistance of an element is $dR = \rho\dfrac{dr}{A}$,

where the cross-sectional area through which current flows, $A = 2\pi rL$.
This is the curved surface area of an element of thickness *dr*.

Hence, $dR = \rho \dfrac{dr}{2\pi rL}$.

So the total resistance across the entire thickness of the hollow cylinder is

$$R = \int_{r_1}^{r_2} dR = \frac{\rho}{2\pi L}\int_{r_1}^{r_2} \frac{1}{r}\,dr = \frac{\rho}{2\pi L}\ln\left(\frac{r_2}{r_1}\right) \quad \text{(shown)}$$

C3 **(a)**

$$\frac{1}{R_{CD}} = \frac{1}{R} + \frac{1}{r}$$

$$R_{CD} = \frac{rR}{R+r}$$

$$R_{AB} = R_{CD} + 2r = \frac{rR}{R+r} + 2r = \frac{rR + 2r(R+r)}{R+r}$$

$$R_{AB} = \frac{3rR + 2r^2}{R+r}$$

(b) Since the extra segment makes no difference to the overall resistance, set R_{AB} = R_{CD} = R

$$R_{AB} = \frac{3rR + 2r^2}{R+r} = R$$

$$3rR + 2r^2 = R(R+r)$$

$$R^2 - 2rR - 2r^2 = 0$$

Solving for *R*,

$$R = \frac{-(-2r) \pm \sqrt{(-2r)^2 - 4(1)(-2r^2)}}{2(1)}$$

$$= \frac{2r \pm \sqrt{12r^2}}{2}$$

$$= \frac{2r \pm 2r\sqrt{3}}{2}$$

$$= \left(1+\sqrt{3}\right)r \quad \text{or} \quad \left(1-\sqrt{3}\right)r \text{ (rejected as } R \text{ cannot be negative)}$$

Hence, $R = \left(1+\sqrt{3}\right)r$

C4 **(b)** **(i)** B, D, E **or** C, F, H

(ii)

$$\frac{1}{C_T} = \frac{1}{3C} + \frac{1}{6C} + \frac{1}{3C}$$

$$= \frac{5}{6C}$$

$$C_T = \frac{6}{5}C = 1.2 \times 25 = 30\ \mu\text{F}$$